INTERNATIONAL MATHEMATICAL UNION

FIELDS MEDAL 2026

YU DENG

SHORT CITATION

For his work in partial differential equations, including the rigorous derivation of the Boltzmann equation from hard-sphere dynamics for rarefied gases, the derivation of wave kinetic equations from nonlinear dispersive systems, and probabilistic approaches to nonlinear Schrödinger dynamics.

LONG CITATION

Combining techniques from partial differential equations, probability, and mathematical physics, Yu Deng has made fundamental contributions to the rigorous analysis of complex dynamical systems, clarifying how macroscopic and statistical behavior emerge from microscopic dynamics.

Together with Zaher Hani and Xiao Ma, Deng achieved landmark results on the derivation of the Boltzmann equation from deterministic hard-sphere dynamics in the low-density regime. His work provides a rigorous derivation of Boltzmann's equation for rarefied gases, valid for as long as the corresponding Boltzmann solution exists, overcoming longstanding obstacles in connecting Newtonian particle systems with kinetic theory and addressing a central problem in nonequilibrium statistical mechanics.

In joint work with Hani, Deng made major advances in the derivation of wave kinetic equations from nonlinear dispersive systems. These results explain how statistical behavior and energy transfer arise from deterministic nonlinear dynamics, contributing significantly to the mathematical foundations of wave turbulence theory.

Beyond kinetic limits, together with Andrea Nahmod and Haitian Yue, Deng developed innovative probabilistic approaches to nonlinear Schrödinger equations (such as random averaging operator and random tensor theories), yielding new insights into long-time behavior, propagation of randomness, and stability for nonlinear dispersive partial differential equations. His work has substantially expanded the scope of rigorous analysis in nonlinear dynamics and has had a deep impact on the study of dispersive equations and mathematical physics.